RETRACTIONS OF FAITHFULLY FLAT ETALE MAPS PERSIST AFTER LOCALIZING ALONG A CLOSED SUBSCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the descent of the given etale algebra over the localization to an etale algebra over A, the adjunction of a localization making it faithfully flat so that the hypothesis applies, and the localization of the resulting retraction back to A_Z^tilde.

Editorial note. The label lemma-have-sections-localize names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-have-sections-localize –start 1702 –end 1720 –local-id 28008 returned RESULT=OK (L28008 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfB.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Descent of an etale algebra along the localization map. Algebra, the lemma describing etale algebras over a localization as the base change of etale algebras over the original ring. Supplies the etale ring map A -> B' with C = B' tensor over A with A_Z^tilde.
- Openness of the image of an etale morphism and covers by the image and a basic open. Algebra, the proposition that the image of an etale (more generally flat finitely presented) morphism is open, and the elementary cover by a basic open D(f). Supply the element f in the ideal such that the spectrum of A is the union of the image U' and D(f).
- The localization construction attached to a closed subscheme. Same chapter, the construction of A_Z^tilde together with the fact that Z is a closed subscheme of its spectrum. Supplies the localization along which the obtained retraction is taken.

Source reference. Original reference 096V: sources/stacks-proetale/pinned-original-source.tex, source lines 710–710. Prerequisite pointer invoked at proof line(s) 1706; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a ring such that every faithfully flat étale ring map $A \to B$ has a retraction. Let $Z \subset \mathrm{Spec}(A)$ be a closed subscheme. Let $A \to A^\sim_Z$ be as constructed in Lemma 096V. Then every faithfully flat étale ring map $A^\sim_Z \to C$ has a retraction.*

Proof. There exists an étale ring map $A \to B'$ such that $C = B' \otimes_A A^\sim_Z$ as $A^\sim_Z$-algebras. The image $U' \subset \mathrm{Spec}(A)$ of $\mathrm{Spec}(B') \to \mathrm{Spec}(A)$ is open and contains $V(I)$, hence we can find $f \in I$ such that $\mathrm{Spec}(A) = U' \cup D(f)$. Then $A \to B' \times A_f$ is étale and faithfully flat. By assumption there is a retraction $B' \times A_f \to A$. Localizing we obtain the desired retraction $C \to A^\sim_Z$. $\square$